Assignment # 2

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Assignment # 2

Department of Computer Science and Engineering
Sejong University
Signals and Systems
Course # 005246
Assignment # 2

Credit Hours:	03	**Semester:**	2026-I
Assignment Date:	March 30, 2026	**Total Marks:**	10
Course Instructor:	**Prof. Amad Zafar**	**Submission Date:**	April 6, 2026

Question # 1 **(3 marks)**

Consider a discrete-time system with input $x[n]$ and output $y[n]$. The input-output relationship for this system is.

$$y[n] = x[n]x[n-2]$$

(a) Is the system memoryless? **(1 mark)**
(b) Determine the output of the system when the input is $A\delta[n]$, where A is any real or complex number. **(1 mark)**
(c) Is the system invertible? **(1 mark)**

Question # 2 **(3 marks)**

Consider a continuous-time system with input $x(t)$ and output $y(t)$ related by

$$y(t) = x(\sin(t))$$

(a) Is this system causal? **(1 mark)**
(b) Determine whether the system is linear. **(2 marks)**

Question # 3 **(4 marks)**

For each of the following input-output relationships, determine whether the corresponding system is linear, time invariant, or both.

i) $y[n] = x^2[n-2]$ **(2 marks)**

ii) $y(t) = t^2x(t-1)$ **(2 marks)**

Question # 1 **(3 marks)**

Consider a discrete-time system with input $x[n]$ and output $y[n]$. The input-output relationship for this system is.

$$y[n] = x[n]x[n-2]$$

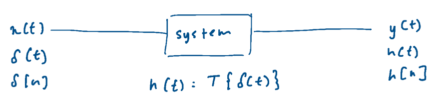
(a) Is the system memoryless? **(1 mark)**
(b) Determine the output of the system when the input is $A\delta[n]$, where A is any real or complex number. **(1 mark)**
(c) Is the system invertible? **(1 mark)**

a) $y[n] = x[n]\, x[n-2]$

system has memory, it depends on past value of time

b) output of system :

$x[n] = A\delta[n]$, A complex number



$h(t) = T\{\delta(t)\}$
$h[n] = T\{\delta[n]\}$

$y[n] = x[n]\, x[n-2]$
$= A\delta[n] \cdot A\delta[n-2]$
$= A^2 \delta[n]\,\delta[n-2]$
$= 0$

c) system is non invertible

Question # 2 CT **(3 marks)**

Consider a continuous-time system with input $x(t)$ and output $y(t)$ related by

$$y(t) = x(\sin(t))$$

(a) Is this system causal? **(1 mark)**
(b) Determine whether the system is linear. **(2 marks)**

a) $y(t) = x(\sin(t))$

$t=0$, $\sin(0) = 0$, $\sin t = t$

$t=1$, $\sin(1) = 0.84$, $\sin t < 1$, past value

$t=-1$, $\sin(-1) = -0.84$, $\sin t > -1$, future value

$\therefore$ system is non-causal

causal - output at curr time depends upon present or past value of input (no future)

anticausal - output at current time depends upon future value of input

non-causal - depends on past AND future

b) linear or not?

$y(t) = x(\sin(t))$

$T\{a_1x_1 + a_2x_2\} = a_1y_1 + a_2y_2$

linear - satisfy additivity and scaling

$y(t) = x(\sin(t))$

$T\{a_1 x_1 + a_2 x_2\} = a_1 y_1 + a_2 y_2$

linear - satisfy additivity and scaling

① additivity
$x_1 \xrightarrow{T} y_1$
$x_2 \xrightarrow{T} y_2$
$x_1 x_2 \xrightarrow{T} y_1 y_2$

② scaling
$x \longrightarrow y$
$\alpha x \longrightarrow \alpha y$

$x(t) = a_1 x_1(\sin t) + a_2 x_2(\sin t) = a_1 y_1 + a_2 y_2$

$a_1 x_1(\sin t) + a_2 x_2(\sin t) = a_1 x_1(\sin(t)) + a_2 x_2(\sin(t))$

① satisfies additivity

$x(t) \rightarrow ax\,t$ $\quad t = \sin t$

$y(t) = (ax)(\sin t)$
$= (ax)\,t$
$= x(t)$

② satisfies homogenity

$\therefore$ system is linear

Question # 3 **(4 marks)**

For each of the following input-output relationships, determine whether the corresponding system is linear, time invariant, or both.

i) $y[n] = x^2[n-2]$ **(2 marks)**

ii) $y(t) = t^2 x(t-1)$ **(2 marks)**

check for linear, time invariant or both

CT $\rightarrow$ shift invariance
DT $\rightarrow$ time invariance

i) $y[n] = x^2[n-2]$

additivity

$y = x^2$
$x_1 \xrightarrow{T} y_1 = x_1^2$
$x_2 \xrightarrow{T} y_2 = y_2^2$
$x_1 + x_2 \xrightarrow{T} y_3 = (x_1 + x_2)^2$
$= x_1^2 + x_2^2 + 2x_1 x_2$
$\neq y_1 + y_2$

$\therefore$ not linear

time invariant

$x(t \pm \tau) \xrightarrow{T} y(t \pm \tau)$

$y[n] = x^2[n-2]$

i/p $\quad x^2[n-2-\tau] \rightarrow y[n-\tau]$

o/p $\quad y[n-\tau] = x^2[n-2-\tau]$

$\therefore$ Time invariant

$\rightarrow$ system is time invariance only

ii) $y(t) = t^2 x(t-1)$

$x_1 + x_2 \xrightarrow{T} y_1 + y_2$

$t^2 x_1(t-1) + t^2 x_2(t-1)$

$= t^2[x_1(t-1) + x_2(t-1)] = y_3$

$x_1 \xrightarrow{T} y_1 = t^2 x_1(t-1)$

$x_2 \xrightarrow{T} y_2 = t^2 x_2(t-1)$

$x_1 + x_2 = y_3 = t^2 x_1(t-1) + t^2 x_2(t-1)$
$= t^2[x_1(t-1) + x_2(t-1)]$

✓ additivity satisfies

$x(t) \rightarrow ax(t)$

$y(t) = t^2 ax(t-1)$
$= at^2 x(t-1)$
$= a(x(t))$

✓ scaling satisfies

$\therefore$ linear

Time invariant

✓ additivity satisfies

Time invariant

$x(t \pm \tau) \xrightarrow{T} y(t \pm \tau)$

$\rightarrow$ system is linear but not time invariant

i/p $y(t) = t^2 x(t-1)$

$= t^2 x(t-1-\tau)$

o/p $y(t-\tau) = (t-\tau)^2 t^2 x(t-1-\tau)$

$\therefore$ Time invariant not satisfied